

Advanced Natural Language Processing

Discussion 5: Towards Transformers

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Master of AI / DSDM
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Maastricht University

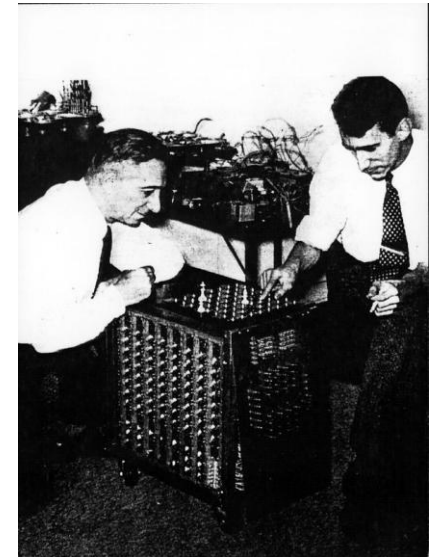
Discussion Lecture: History of Deep Learning



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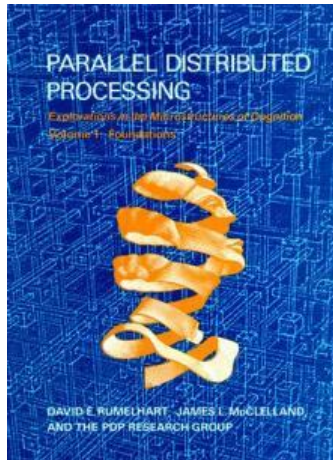
Claude Shannon's Chess computers & Maze Robots

- Shannon known for information theory (1948), which gave us entropy, bits, and the mathematical idea of information.
- Shannon also explored learning machines, neural nets, and robotics — long before AI became a discipline.
- Shannon loved randomness. He believed that randomness + selection can produce intelligence.



Concept (Shannon)	Connection to modern deep learning
Entropy	loss functions, uncertainty estimation
Information channel capacity	limits of model compression and generalization
Encoding/decoding theory	autoencoders, representation learning

1988: Back-propagation: another AI (neural) summer



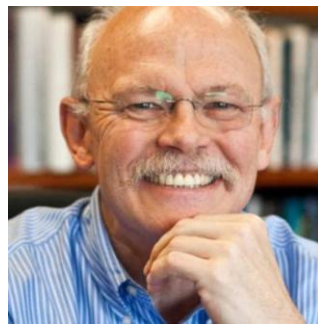
- 3-layer networks
- Could learn from data
- Also capable of capturing non-linear relations

Progress in optical character recognition and other simple pattern recognition tasks.

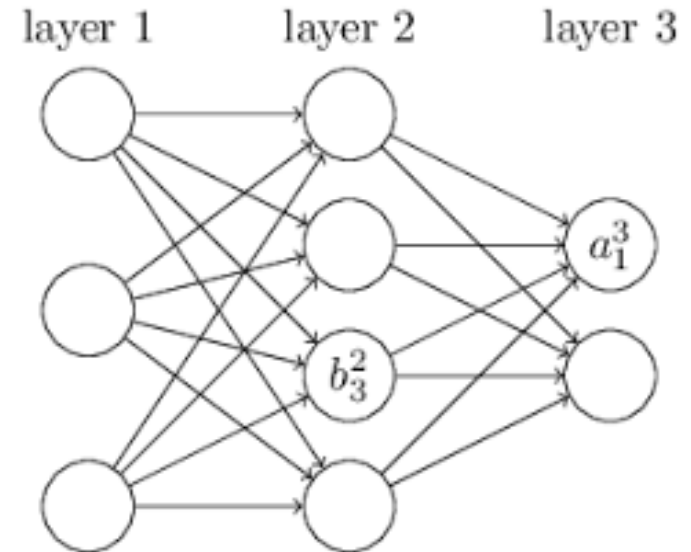
The sky seemed to be the limit

...

D. Rumelhart



J. McClelland

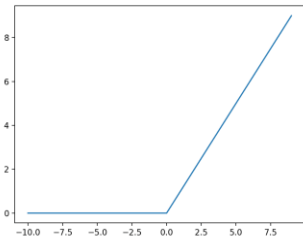


See also:

- Parallel Distributed Processing. Volume 1: Foundations
- D. Rumelhart, J. McClelland, and PDP Research Group (Eds.). . MIT Press, Cambridge, MA, (1986)

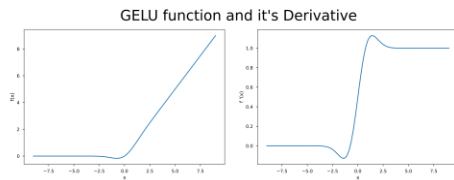
Activation Functions

- Hidden layers traditionally used a sigmoid function, but today is more varied, with rectifier (ReLU-rectified linear unit) being common.



$$f(x) = x^+ = \max(0, x),$$

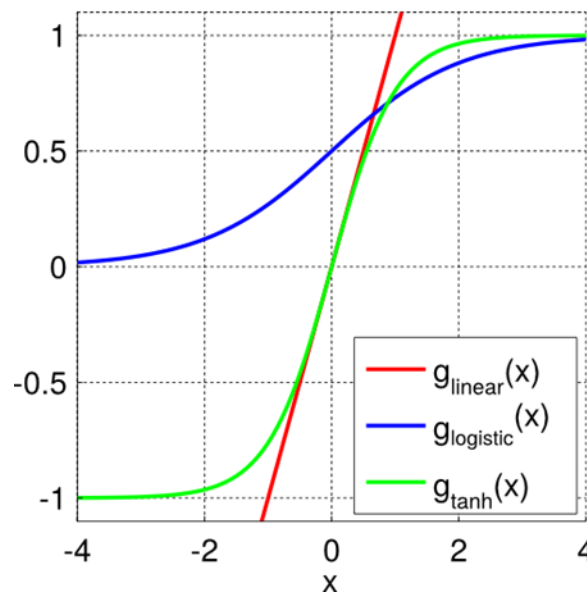
ReLU



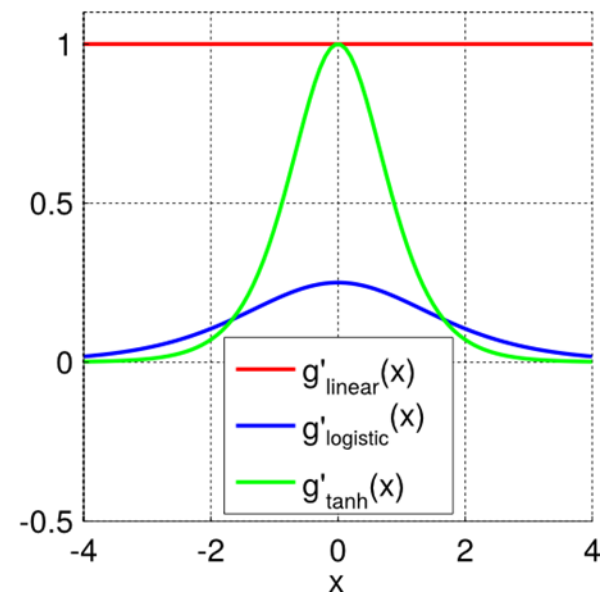
$$\text{gelu}(x) = \frac{1}{2}x(1 + \text{erf}(\frac{x}{\sqrt{2}}))$$

GELU

Some Common Activation Functions



Activation Function Derivatives



$$g_{\text{linear}}(z) = z$$

$$g_{\text{logistic}}(z) = \frac{1}{1+e^{-z}}$$

$$g_{\text{tanh}}(z) = \frac{\sinh(z)}{\cosh(z)} = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$



Softmax versus Sigmoid

The softmax function and the sigmoid function are similar. The softmax operates on a vector while the sigmoid takes a scalar.

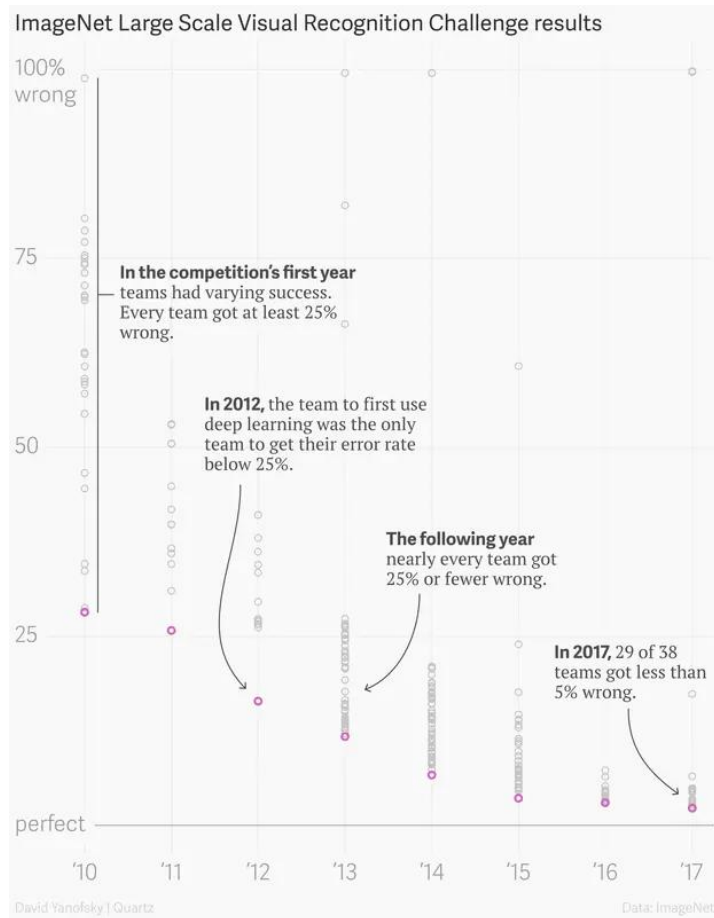
- Single classification (scalar): sigmoid.
- Multi-class (vector): softmax.

$$\text{Softmax } \sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

$$\text{Sigmoid } S(x) = \frac{1}{1 + e^{-x}}$$

for two classes, softmax = sigmoid (you can try this yourself).

2012: Deep learning breakthrough

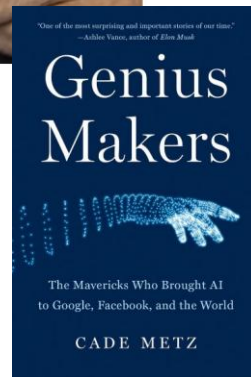


Yoshua Bengio, Geoffrey Hinton, Yann LeCun share 'Nobel of computer science' for work on deep learning

[Emily Chung](#) · CBC News · Posted: Mar 27, 2019 6:00 AM ET | Last Updated: March 27, 2019

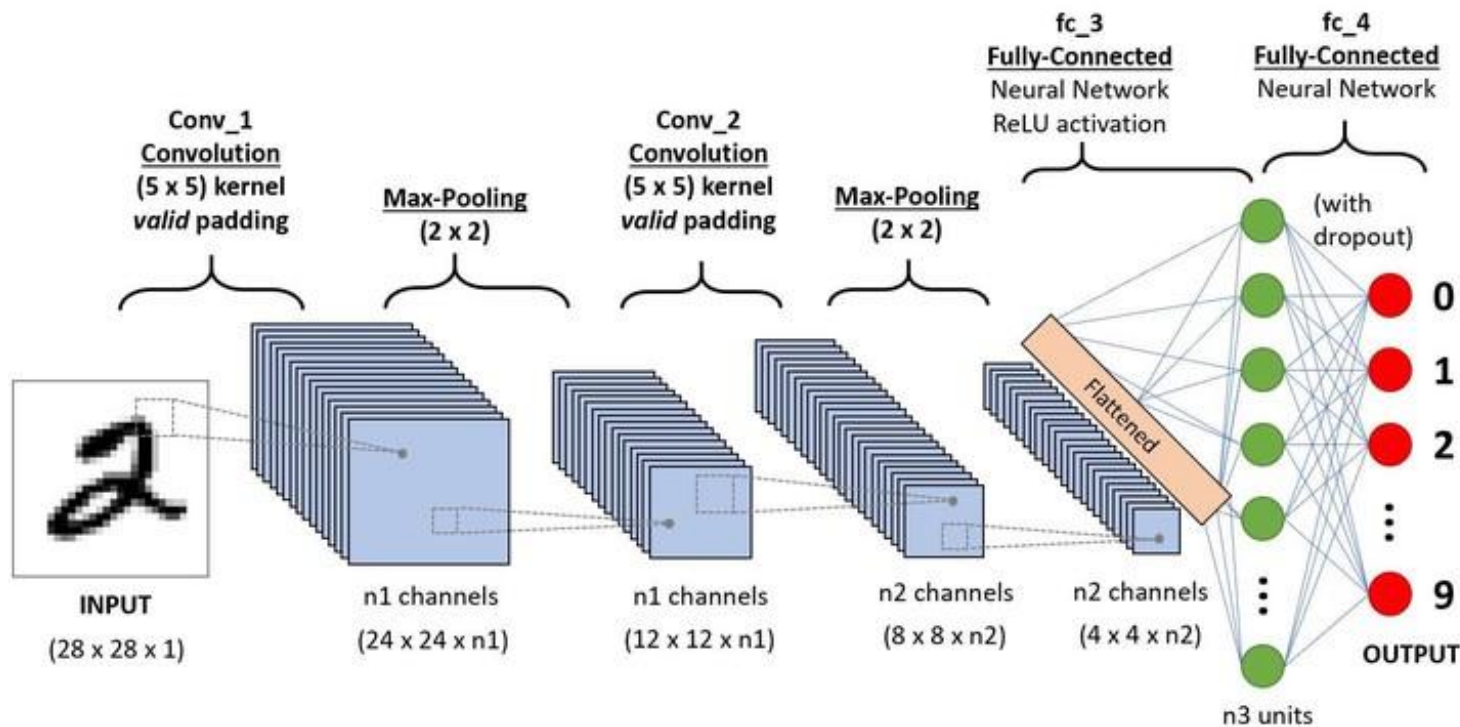


In 2012, Geoffrey Hinton found an algorithm to train multiple layers (deep) with billions of neurons and used it to outperform every other algorithm in image recognition and reached above human recognition levels quickly after that.



2012: Deep Learning for computer vision

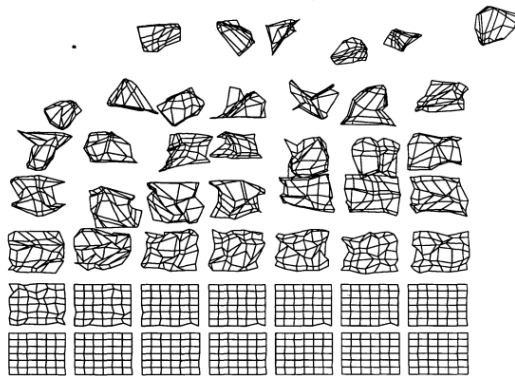
- Multi-layer (>3, often 10+) networks can now be trained using convolution.
- Breakthrough in Computer Vision tasks.



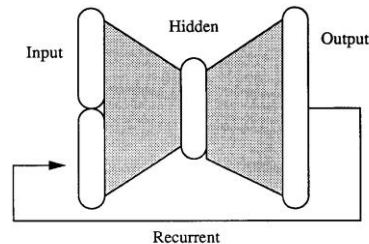
Padding preserves size of image (each layer conv makes it smaller), Stride deals with convolution on edges, otherwise middle gets more conv. Pooling (max=take most prominent pixels or average = preserves overall picture) reduces spatial size to reduce complexity and computational cost. Softmax at end for multi-class classification.

1990-1993: first attempts with back propagation, RNN and self-organizing maps.

Neural Networks in Natural Language Processing and Information Retrieval

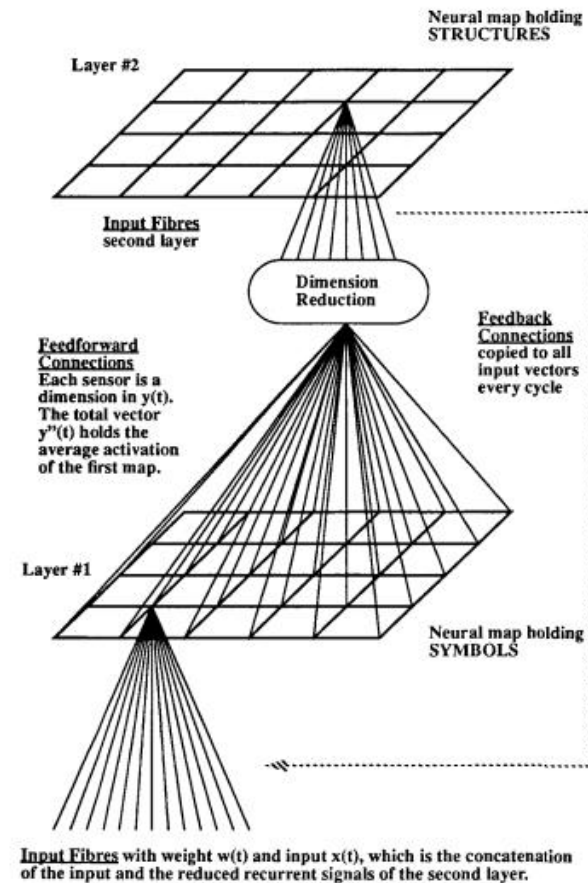


Johannes Cornelis Scholtes

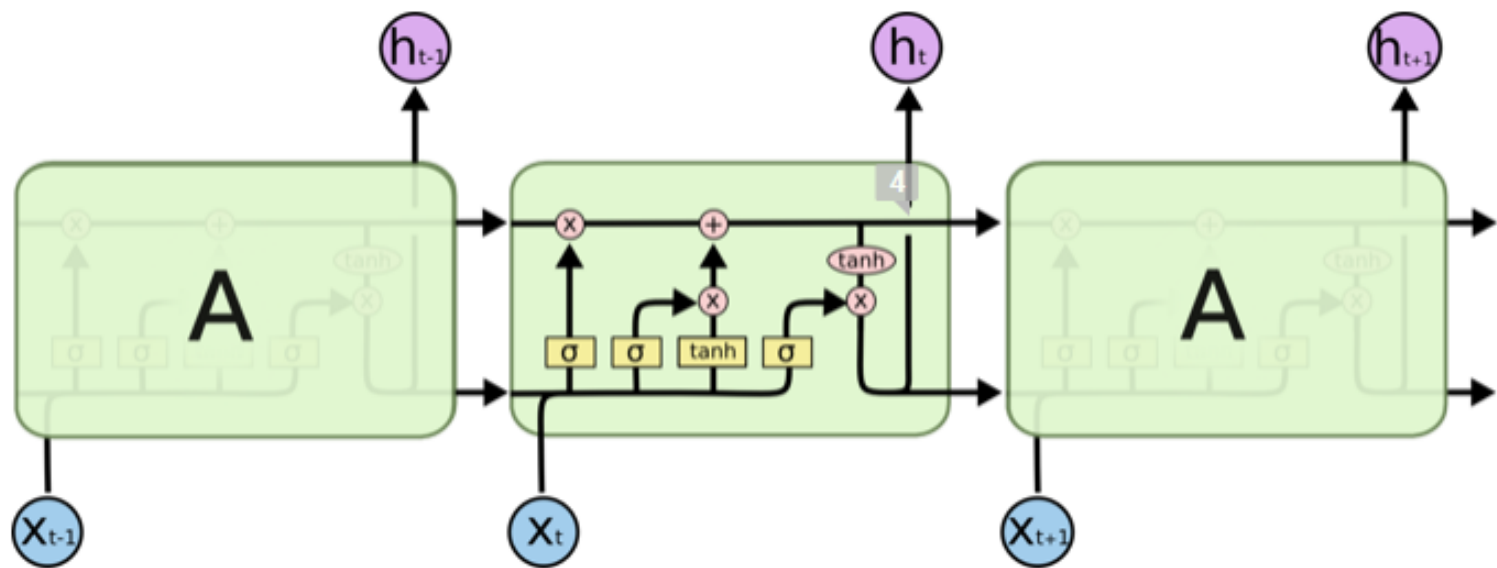


Challenges:

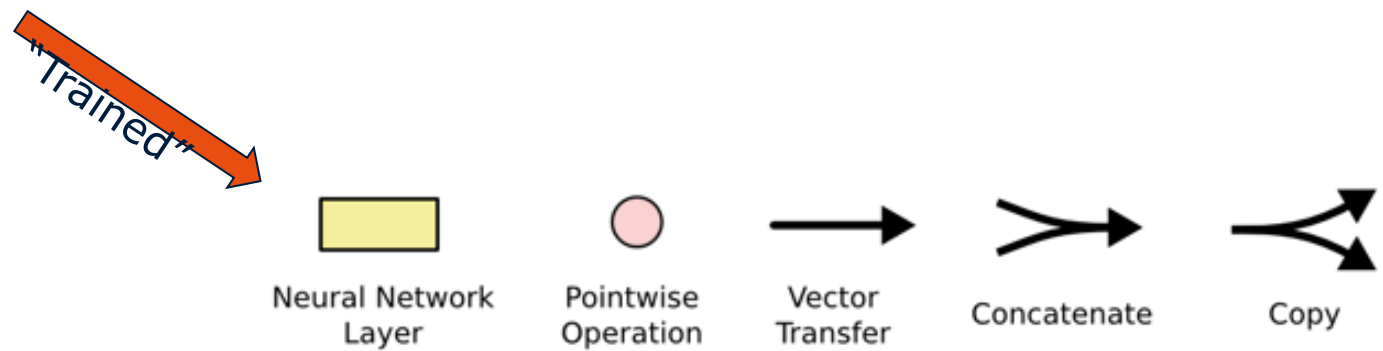
- Not enough data
- Not enough CPU power
- Not enough memory
- Not able to get multi-layer training stable with then known algorithms.



LSTMs also have this chain like structure, but the repeating module has a different structure. Instead of having a single neural network layer, there are four layers (σ , σ , $\tanh$, and σ) interacting in a very special way.



The repeating module in an LSTM contains four interacting layers.



Questions on the lecture?

Now let's form groups of 4



Discuss Exercises Lecture

Exercise 1

Review the transition from the McCulloch-Pitts neuron (1943) and Rosenblatt's Perceptron (1958) to the formal proof of linear separability limitations published by Minsky and Papert in 1969.

- State the fundamental mathematical differences between a McCulloch-Pitts neuron and a Rosenblatt Perceptron regarding input values, individual input weights, and the threshold activation function.
- The single-layer Perceptron uses the decision rule:
- $y = 1 \Leftrightarrow w_0 + \sum_{i=1}^n w_i x_i \geq 0$
- Write out the system of linear inequalities for the 2-input XOR logic gate ($x_1, x_2 \in \{0,1\}$). Show algebraically why no combination of weights (w_0, w_1, w_2) can satisfy all four conditions simultaneously.
- Why did Minsky and Papert's demonstration that single-layer perceptrons cannot solve non-linear problems trigger the first "AI Winter", and what architectural addition in the 1986 *Parallel Distributed Processing* (PDP) framework finally overcame this limitation?

Exercise 2: Backpropagation, Loss Functions, and Gradient Computations

Modern neural architectures rely on the backpropagation training loop popularized by Rumelhart, Hinton, and Williams (1986).

- Contrast the usage of **Mean Squared Error (MSE)** and **Categorical Cross-Entropy**:
 - Which loss function is suited for continuous regression, and which is designed for discrete classification?
 - Why is the combination of a **Softmax** output layer and Cross-Entropy loss preferred over MSE for multi-class classification?

- Using the chain rule derivation from the lecture:

$$\frac{\partial E}{\partial w_{ij}} = \frac{\partial E}{\partial o_j} \cdot \frac{\partial o_j}{\partial \text{net}_j} \cdot \frac{\partial \text{net}_j}{\partial w_{ij}} = o_i \delta_j$$

Given a single output neuron j using a standard sigmoid activation $\varphi(z) = \frac{1}{1+e^{-z}}$ (where $\varphi'(z) = o_j(1 - o_j)$) and squared error loss $E = \frac{1}{2}(o_j - t_j)^2$, derive the exact expression for δ_j .

- Explain how error blame δ_j is backpropagated to a hidden neuron from the subsequent layer k . Why do saturating activations like sigmoid or tanh cause the vanishing gradient problem in deep multi-layer networks, and how do modern activations like ReLU or GELU help mitigate this?

Exercise 3: Spatial vs. Temporal Modeling (CNNs to RNNs & LSTMs)

Trace the shift from spatial deep learning (AlexNet, 2012) to sequential deep learning models designed for natural language.

- Convolutional Neural Networks (CNNs) achieved superhuman performance on ImageNet using spatial operations (convolution, stride, padding, pooling). Explain why applying 1D CNNs or dilated convolutions to text struggled to serve as a complete solution for modeling natural language syntax and semantics.
- Standard Recurrent Neural Networks (RNNs) pass a hidden state vector h_t through time:
 - How does an RNN theoretically achieve Turing completeness on arbitrary input sequences?
 - When training an RNN using Backpropagation Through Time (BPTT), why does repeatedly multiplying hidden state weight matrices across long sentences cause vanishing or exploding gradients?
- Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory (LSTM) network to solve this degradation. How does the internal gating mechanism ($\sigma, \sigma, \tanh, \sigma$) and cell state highway (C_t) allow an LSTM to retain long-term dependencies (e.g., matching "*I grew up in France*" with "*I speak fluent French*") far better than a vanilla RNN? Why did LSTMs still remain bottlenecked by computational complexity and sequential training constraints?

Discussion Lecture: History of Deep Learning



On the Illusion of Biology: Artificial Neurons vs. Real Brains

In 1958, the New York Times reported that the Perceptron was an embryo of an electronic computer expected to 'think', 'grow wiser', and even become 'conscious'. Yet beneath that rhetoric, Rosenblatt's device was simply a linear classifier summing weighted inputs with a step threshold.

Today, we build deep neural networks with billions of parameters, but the fundamental forward-pass unit remains a weighted sum followed by an activation function.

Are modern deep learning systems actually modeling the biological cognition of the brain as early cyberneticists envisioned, or did we just repurpose neurobiological terminology to make multi-variable calculus and linear algebra sound intelligent? If an artificial neuron shares almost nothing with a biological cell except its name, why did the brain metaphor remain so resilient across eighty years of AI development?



On the Linear Separability Trauma: The Cost of Discarding Promising Paradigms

In 1969 Minsky and Papert proved that a single-layer perceptron cannot compute XOR. Funding dried up and neural network research entered a winter that lasted more than a decade, although the way out (hidden layers trained by the chain rule) was mathematically within reach the whole time.

Did the field abandon connectionism because the limitation was insurmountable, or because a rival school convinced the community that it was? What decides which critiques kill a paradigm and which are absorbed by it?

Today deep learning has its own well-documented failures: hallucination, lack of grounding, weak reasoning. Are we at risk of prematurely declaring another paradigm dead, or of refusing to declare this one dead because too much depends on it?

On Depth: When Does More Become Different?

The breakthrough of 2012 added no new operation. It stacked the same one, a weighted sum and a nonlinearity, many more times than before, and called the result “deep”.

When does more of the same become something different? A single perceptron draws a line; two layers draw any shape; a hundred layers recognise a face. Is that a change in kind, or only in quantity, and does the distinction matter?

If understanding can emerge from repetition of a trivial step, what, if anything, is left of the idea that understanding requires a special ingredient? And if nothing emerges, what are we looking at when a deep network succeeds?

Questions?



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Break

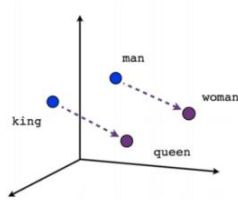
Discussion: Lecture Transfer Learning, Attention, Encoder-Decoder



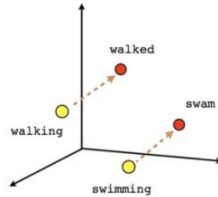
Central Idea to Word Embeddings

- How can we assign an (high-dimensional) encoding to a token that represents morphological, syntactical, semantical and other linguistic properties of a token so that similar linguistic properties result in similar encodings?
- As a result, we should benefit from better linguistic representations and potentially transfer learning properties.

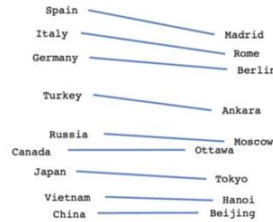
Transfer Learning through Word Embeddings



Male-Female



Verb tense



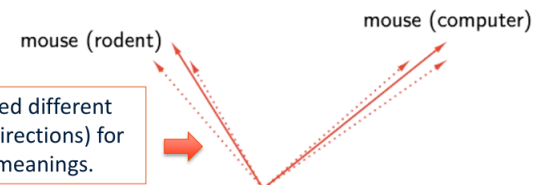
Country-Capital

Context Sensitive Embeddings

Sentences

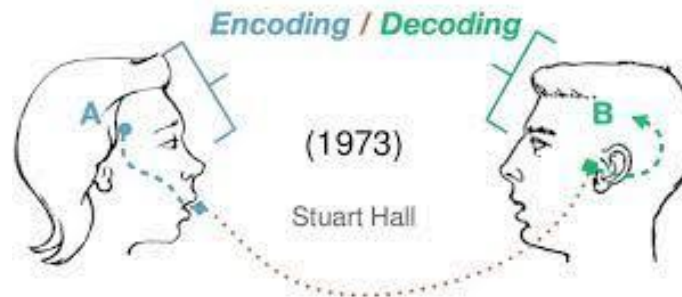
"bank"
"he eventually sold the shares back to the bank at a premium."
"the bank strongly resisted cutting interest rates."
"the bank will supply and buy back foreign currency."
"the bank is pressing us for repayment of the loan."
"the bank left its lending rates unchanged."
"the river flowed over the bank."
"tall, luxuriant plants grew along the river bank."
"his soldiers were arrayed along the river bank."
"wild flowers adorned the river bank."
"two fox cubs romped playfully on the river bank."
"the jewels were kept in a bank vault."
"you can stow your jewellery away in the bank."
"most of the money was in storage in bank vaults."
"the diamonds are shut away in a bank vault somewhere."
"thieves broke into the bank vault."
"can I bank on your support?"
"you can bank on him to hand you a reasonable bill for your services."
"don't bank on your friends to help you out of trouble."
"you can bank on me when you need money."
"i bank on your help."

The different meanings of the word "bank" will get very different vectors, depending on this context. So, the "financial bank" will be closer to other financial words than the "river bank". This is essential for good language understanding!



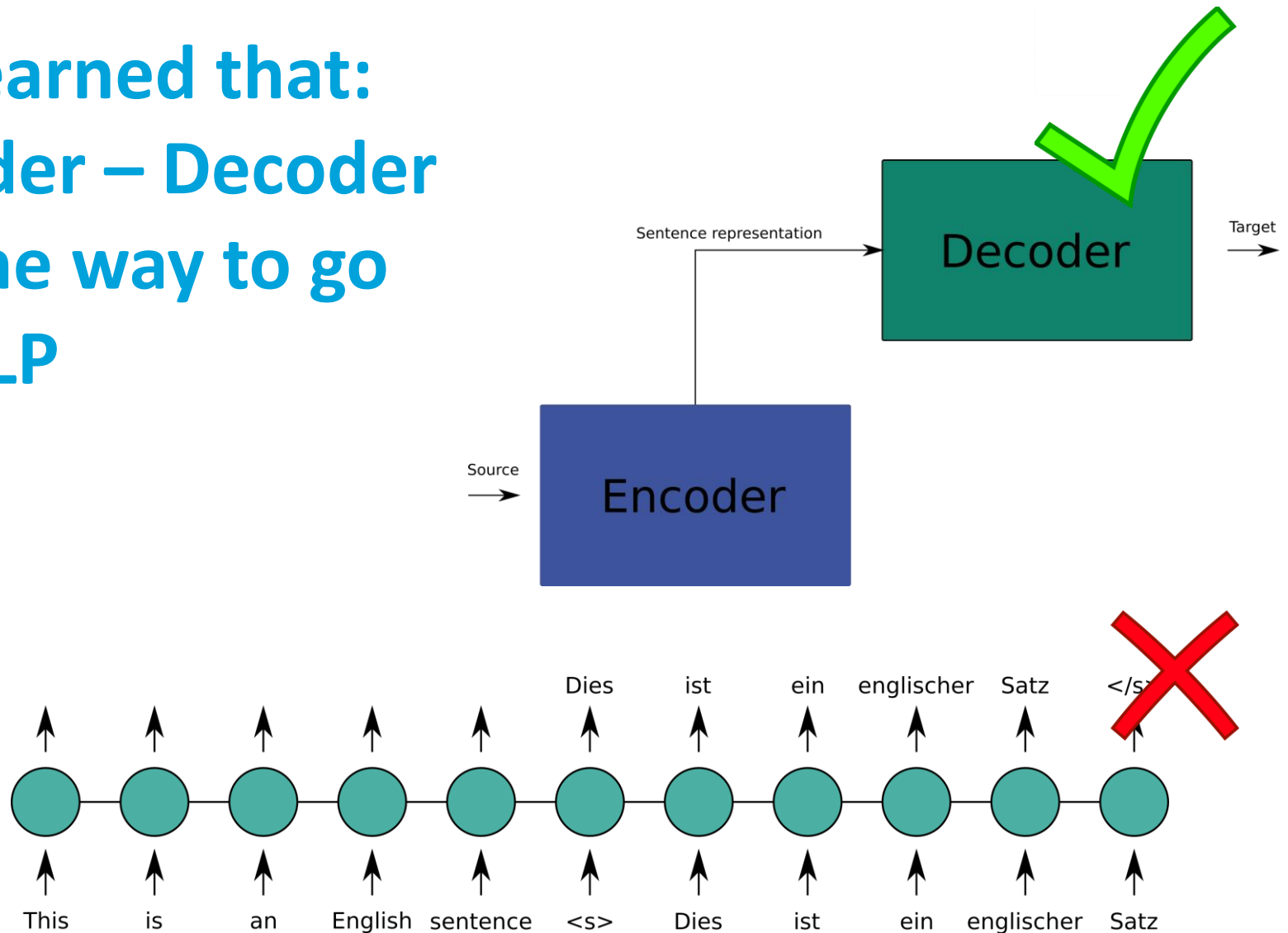
Encoding - Decoding

- Understanding language is an iterative process for humans: When we read or hear a sentence, we process each word accumulating information up to the end of the text. This can be seen as an encoding process.



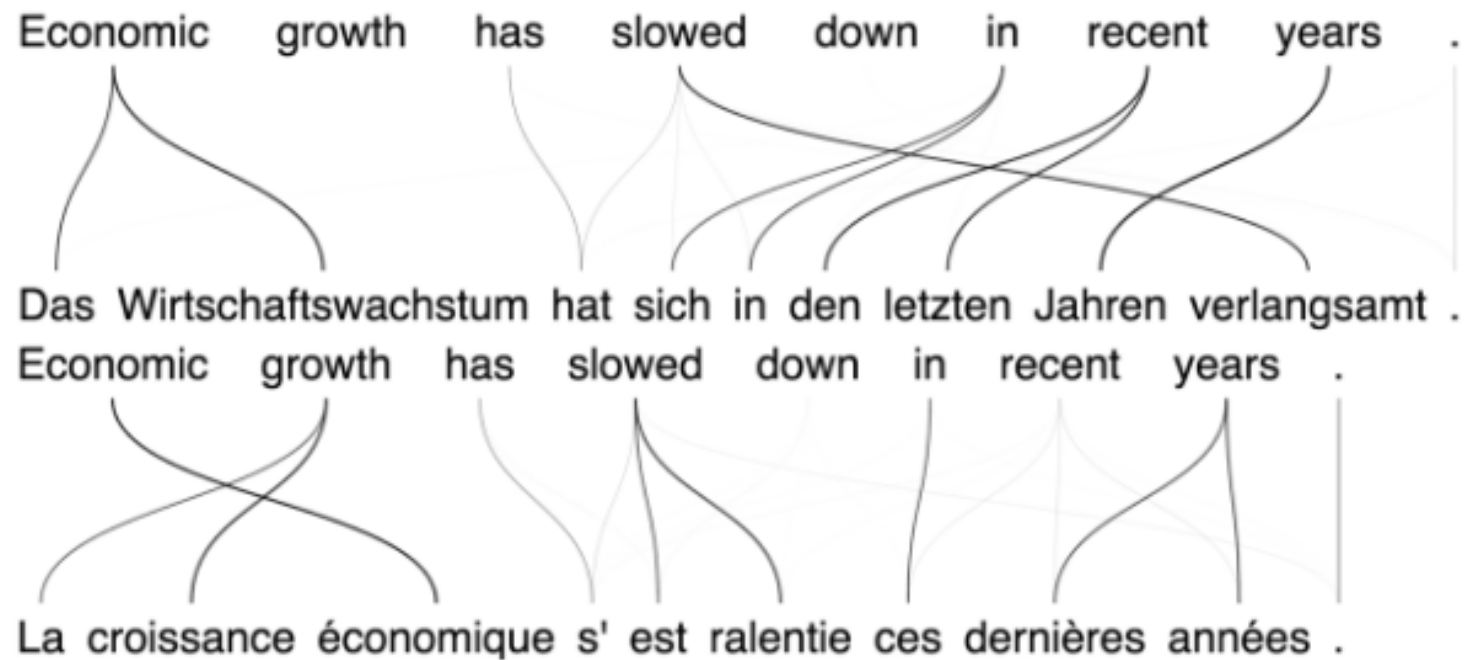
- Then, we perform a certain task: translation, Q&A, abstracting, story telling, ... These can all be seen as decoding processes.

We learned that:
Encoder – Decoder
are the way to go
for NLP



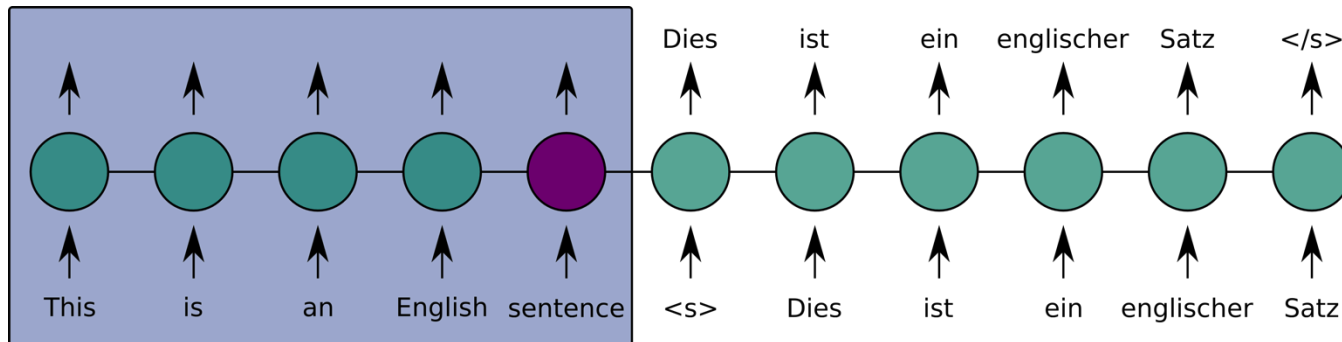
Attention to Deal with Alignment

- Idea:
 - Focus on different parts of the source sentence
 - Different parts are important for different target words
 - Alignment in SMT



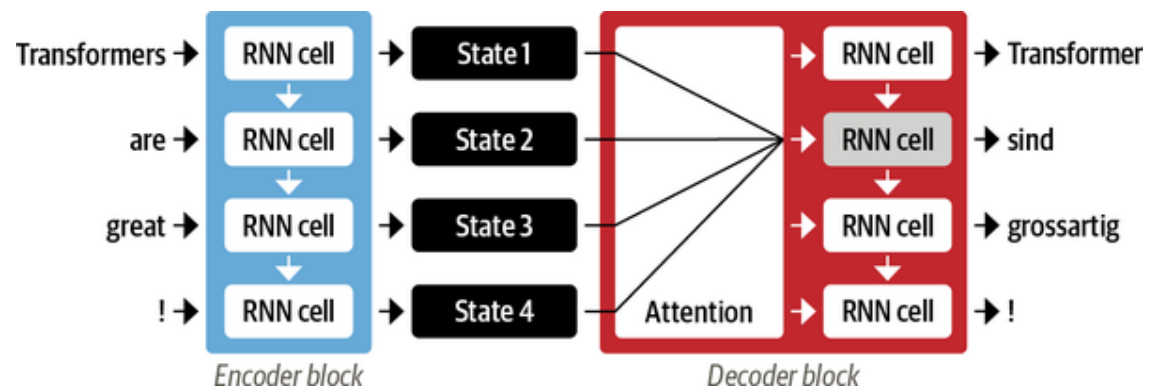
Motivation

- Problem:
 - Bottleneck context vector
 - Fixed size representation of source language



Attention Mechanisms

- Attention: instead of producing one hidden state for the entire input sequence, the encoder outputs a hidden state at each step.
- In order to prevent too much input: the decoder assigns a different weight (e.g. **attention**) to the encoder states at every decoding timestep.
- By focusing on which input tokens are most relevant at each timestep, these attention-based models can learn nontrivial **alignments and relations** between the words in a generated translation and those in a source sentence.



Teaching note: This is the general recipe used everywhere attention appears later in the course (self-attention, multi-head attention, Transformers): many states in, one weighted summary out.

Attention to look up values from ambiguous data

- Use current element to query data based
- Find similar keys in database
 - Importance:
 - Similarity between query and key
- Select associated values
- Example: How expensive is a burger?

Query: Burger

Key	Value
Cheeseburger	12
Fries	5
Hamburger	10
Ice cream	6

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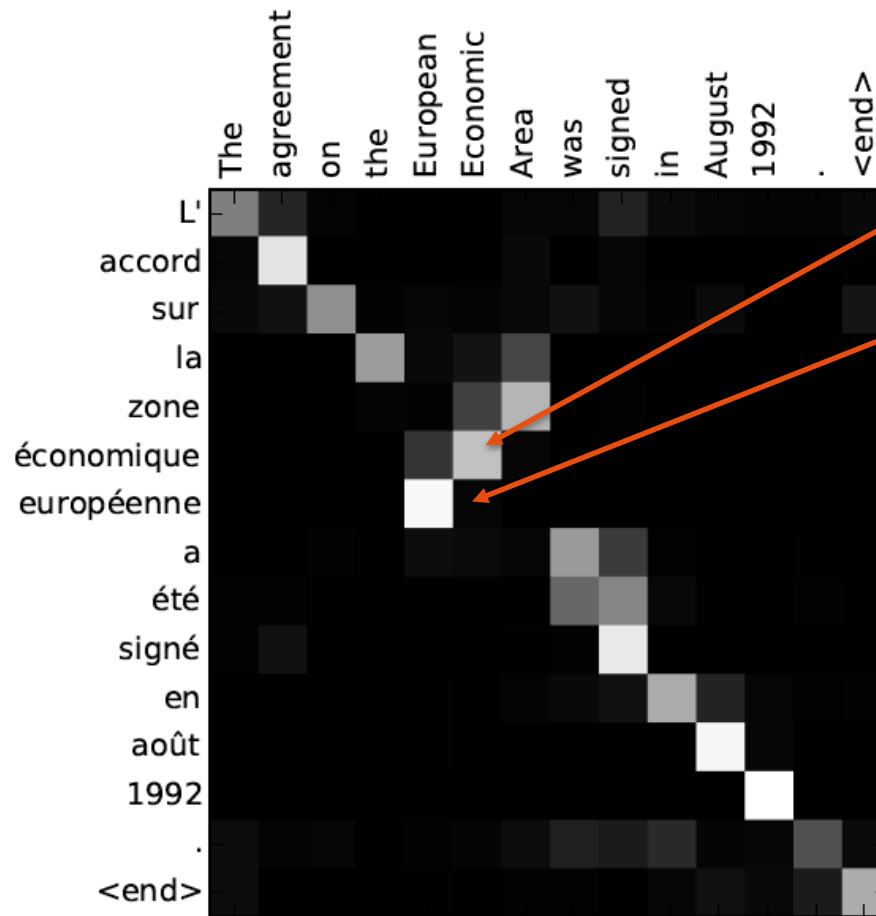
Key	Value	Similarity
Cheeseburger	12	0.5
Fries	5	0
Hamburger	10	0.5
Ice cream	6	0

Teaching note: This burger example IS attention: query="burger", keys=menu items, values=prices, weights=similarity — exactly Q/K/V in the next slides, just without the matrix notation yet.

Average of sum of the value
times (x) similarity (aka weights)



RNN Encoder-Decoder with Attention



Teaching note: Off-diagonal bright cells = the model reordering words on its own — the alignment problem greedy translation couldn't solve (slide 14).

- Light is HIGH attention
- Dark is lower attention
- Black is no attention
- See how attention solves the alignment problem!

Questions on the lecture?

Discuss Exercises Lecture

Exercise 1: Decoding Strategies & The Bottleneck Problem (Greedy vs. Beam Search)

The lecture illustrates the limitations of standard RNN sequence-to-sequence generation and the distinction between greedy decoding and beam search. Without attention, BLEU scores collapse as sentence length increases beyond 20–30 words.

- Explain why a naive sequence-to-sequence model using a single fixed-dimension vector c (the final hidden state of the encoder) creates an information "bottleneck". What specific performance failure occurs on long sentences (e.g., 50 words) compared to short sentences?
- Contrast **Greedy Search** and **Beam Search** during decoding:
 - How does greedy decoding choose tokens at each target position, and why can this lead to suboptimal global sentence translations (e.g., "Yoda-like" phrasing)?
 - How does beam search with a beam width of $k \in [5, 10]$ mitigate this issue?
- Relate beam search decoding back to the **Label Bias Problem** discussed with MEMMs in Lecture 6. Why does optimizing path probabilities across a candidate beam prevent the decoder from being locked into locally high-probability errors?

Exercise 2: Computing Bahdanau Attention Weights & Context Vectors

The lecture shows attention as an information retrieval lookup over keys, values, and queries. Slide 33 details the mathematical computation of attention weights $\alpha_{t,i}$ and the dynamic context vector context_t . Suppose a decoder at time step $t = 1$ has generated a query vector q_1 from its previous state s_0 . The source sentence has 3 encoder states with corresponding key vectors k_1, k_2, k_3 and value vectors v_1, v_2, v_3 .

- Raw alignment scores: $\text{score}(q_1, k_1) = 2.0$, $\text{score}(q_1, k_2) = 1.0$, $\text{score}(q_1, k_3) = 0.0$
- Value representations: $v_1 = [1.0, 0.0]^T$, $v_2 = [0.0, 2.0]^T$, $v_3 = [1.0, 1.0]^T$

Questions:

- Using the softmax function $\alpha_{t,i} = \frac{\exp(\text{score}(q_t, k_i))}{\sum_j \exp(\text{score}(q_t, k_j))}$, compute the attention weights $\alpha_{1,1}$, $\alpha_{1,2}$, and $\alpha_{1,3}$ (round to two decimal places; use $e^2 \approx 7.39$, $e^1 \approx 2.72$, $e^0 = 1.0$).
- Compute the resulting dynamic context vector $\text{context}_1 = \sum_{i=1}^3 \alpha_{1,i} v_i$.
- How does this dynamically computed context vector differ from the fixed memory vector in a standard RNN encoder-decoder model? Explain how error backpropagation flows through this weighted sum into the projection matrices W_Q, W_K, W_V during training.

Exercise 3: Word Alignment & Attention Matrix Interpretation

The lecture display attention heatmaps for English-to-French translation (e.g., *"The agreement on the European Economic Area was signed in August 1992"* → *"L'accord sur la zone économique européenne a été signé en août 1992"*).

- In the attention heatmap, what do the light squares versus the dark/black squares represent in terms of the numerical values of $\alpha_{t,i}$?
- Looking at the phrase *"European Economic Area"* translating to *"la zone économique européenne"*, observe how the attention weights deviate from a simple diagonal line. How does this demonstrate that attention directly solves the classical Machine Translation **alignment problem** and handles syntactic word-order permutations between languages?
- Why does the ability of the decoder to inspect all encoder hidden states at every generation step pave the way toward replacing recurrent connections entirely (leading to the **Transformer** architecture)?

Discussion



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Teaching note: Use today's ideas (the bottleneck problem, soft attention as lookup, word alignment) as concrete cases for the questions ahead — there's no single correct answer to these.

On the Bottleneck: Does a Sentence Have a Meaning That Can Be Stored?

The classic encoder-decoder assumes that the meaning of a sentence can be stored: read the words, produce one vector, hand it over. Attention gave that up. The decoder no longer receives a meaning; it looks back at the words, again and again, whenever it needs them.

Does a sentence have a meaning that exists apart from its words, something that could in principle be held in one place? Or is sentence meaning a process of relating the parts, which exists only while someone (or something) is doing the relating?

If attention works better precisely because it stopped trying to store the meaning, what does that tell us about the project of representing what a sentence means?

On Softness: Do Minds Blend, or Do Gradients Need Them To?

Attention never chooses. It blends every word in the sentence with a weight, because a choice (an argmax) has no gradient and the model could not be trained through it. Everything in a neural network is soft for the same reason.

Human attention does seem to choose: we read one word, hear one voice, miss the gorilla. When we borrow the word “attention” for a weighted sum, are we describing cognition, or describing what gradient descent allows us to build and calling it cognition?

If a theory of the mind has to be differentiable before we can test it on a computer, what kinds of minds can that method never discover?

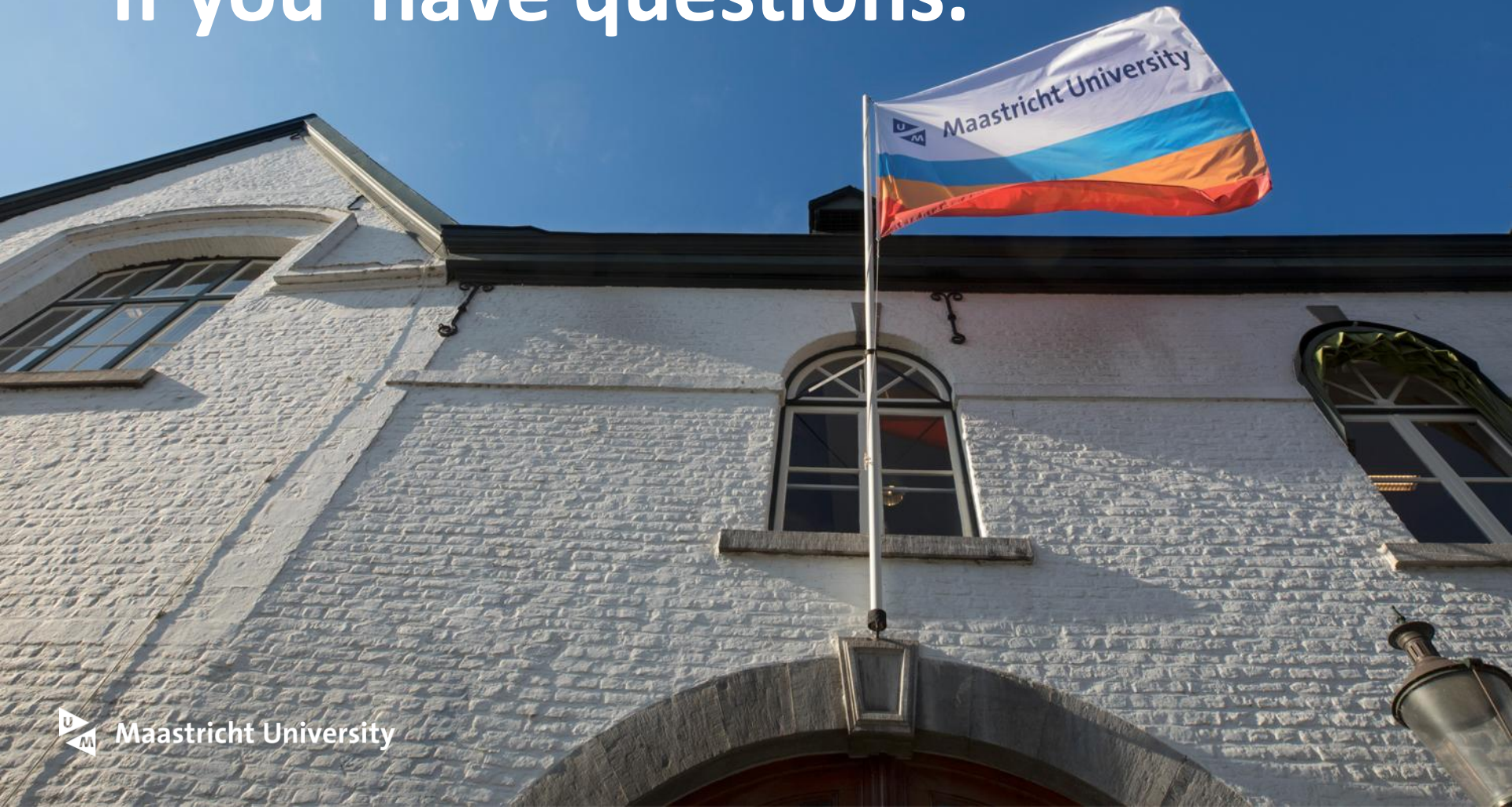
On Translation: What Is It That Stays the Same?

An attention heatmap is a theory of correspondence: it says which English words “are” which French words. The theory was learned from data, with no rules about either language, and it is often right.

Translation assumes that something stays the same when the words change. What is it? The heatmap suggests it is a set of word-to-word links; a translator would say it is a thought, an effect on the reader, or a purpose. When the heatmap shows no clean alignment (idioms, politeness forms, a joke), is that a failure of the model or a fact about languages?

Quine argued that translation is in principle underdetermined: several incompatible translation manuals can fit all the evidence. If so, what is the attention matrix an approximation of, and what does a BLEU score measure?

Tutorials Grammars and
Statistics – We can revit these
if you have questions.



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Tutorials PyTorch, Transformers



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Questions?



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